

Instructions and Expectation

This is a 24-48 hour take-home exercise. We are not looking for a production system or a state-of-the-art model.

We care about:

- how you formulate the problem;
- how you inspect and clean geospatial data;
- whether you establish sensible baselines;
- whether your evaluation is scientifically valid;
- how you reason about uncertainty and failure cases;
- whether your code and explanation are understandable.

You may use open-source libraries and pretrained models. Please document any external models, data, or code you use.

A good submission normally contains:

1. A reproducible notebook or small codebase.
2. A README describing the problem understanding and approach.
3. At least one simple baseline and considered approach.
4. Quantitative evaluation.
5. Visualisation of representative successes and failures.
6. A short discussion of what you would try next.

Do not spend substantial time tuning hyperparameters.

Case study: SAR-Assisted Understanding

Satellite optical imagery is intuitive to interpret but frequently obscured by clouds. SAR imagery can observe the surface under many conditions where optical imagery is unavailable. Your goal is to investigate whether SAR provides useful information when optical observations are partially unavailable.

Dataset

Use a manageable subset of paired Sentinel-1 and Sentinel-2 imagery from BigEarthNet. You do not need to use the complete dataset. Select several land-cover classes and construct a subset that can comfortably be processed on your available hardware.

Task

Build a model that predicts land-cover classes under three conditions:

- A.** Optical only: Use Sentinel-2 imagery.
- B.** Degraded optical: Artificially mask portions of Sentinel-2 imagery to simulate missing/cloud-obscured observations.
- C.** Degraded optical + SAR: Use the degraded Sentinel-2 image together with the corresponding Sentinel-1 observation.

Compare performance between B and C.

Minimum baseline

Implement a simple CNN or pretrained image encoder using optical imagery. Then evaluate the same model after artificially masking part of the optical image.

Contribution

Develop one SAR-assisted approach.

Examples include:

- concatenating SAR and optical channels;
- separate SAR and optical encoders followed by feature fusion;
- mapping SAR features into an optical feature space;
- using SAR to predict information missing from the optical representation.

The architecture does not need to be sophisticated.

Evaluation

Report an appropriate classification metric such as accuracy, macro F1, and others.

Break the results down by: optical only, cloudy/degraded optical, cloudy/degraded optical + SAR, different levels of simulated cloud coverage.